higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE (VOCATIONAL)

WELDING
NQF LEVEL 2

12 MARCH 2019

This marking guideline consists of 6 pages.

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QUESTION 1

1.1 G
1.2 J
1.3 K
1.4 E
1.5 A
1.6 D
1.7 B
1.8 C
1.9 F
1.10 H

(10 × 1) **[10]**

QUESTION 2

2.1
- Connect the power supply cable for the welding machine to the main power socket. Do not switch on until you connect the components.
- Connect the earth clamp cable from the negative connection on the welding machine.
- Connect the electrode holder from the positive connection on the welding machine.
- Switch on the main power supply and then the welding machine.
- Set the correct current required by the electrode you are using. (5)

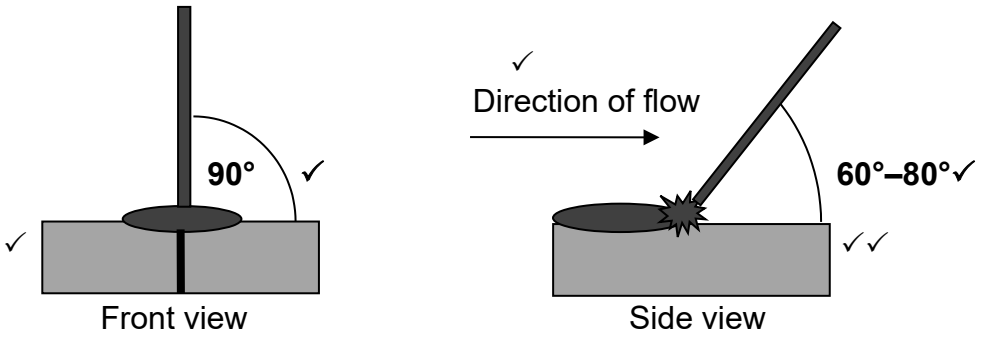
2.2
- Ensure that you are well organised with everything in place.
- Ensure that the area is free from flammable substances that can catch fire when welding.
- Ensure that the area where the welding is to be done is free from dirt and grease.
- Ensure that the area is well lit up to prevent defects in the welding.
- Ensure that the area is shielded properly so that other fellow workers do not get exposed to sparks and arc rays.
- Ensure that there is sufficient space to carry out the welding. (Any 5 × 1) (5)

2.3
- Flammable chemicals must be stored away from heat and open flames.
- Chemicals must be placed on a secured shelve to prevent spillage.
- Chemicals that are reactive with other chemicals must be stored separately.
- The outside of the containers must be cleaned to remove any residues before storing.
- Chemicals must be stored in a properly ventilated room.
- All solvents and chemical must be sealed properly before storing.

(Any 4 × 1) (4)

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2.4



(THREE marks for labels and THREE marks for drawing) (6)

2.5 A – Cracks in the weld
B – Distortion
C – Porosity or slag inclusion
D – Lack of penetration
E – Undercutting (5)
[25]

QUESTION 3

3.1
- Much cheaper than other welding equipment. Examples GMAW, GTAW
- Easily taught
- More portable than other types of welding equipment
- Can also be used to cut material (4)

3.2 A – Welding nozzle
B – Nozzle mixer
C – Acetylene control valve
D – Connection for gas hoses
E – Oxygen control valve (5)

3.3 Acetylene equipment (1)
[10]

QUESTION 4

4.1 4.1.1 Gas cylinder stores different gases.

4.1.2 Gas regulators control the pressure of the gas that is released from the gas cylinders.

4.1.3 Gas hoses transport the gases from the regulators to the gas cutting torch.

4.1.4 Flashback arrestors prevent the flames from entering the gas hoses as well as the gas cylinders.

4.1.5 The cutting torch is used to set a neutral flame so that cutting can be done by the torch.

(5 × 2) (10)

4.2 The artisan is busy with the setting of the three types of flames.✓ He is setting the oxidising flame by allowing the oxygen to mix with the acetylene gas.✓ He sets the final flame known as the neutral flame✓ by pressing the cutting lever until he sees a small blue flame close to the tip of the nozzle.✓ He releases the cutting lever and the torch is set and ready to be used.✓ (5)

4.3

Plate thickness (mm)	Nozzle size	Oxygen pressure (KPA)	Acetylene pressure (KPA)
12	1	**A = 270 kpa**✓	21
20	**B = 2**✓	276	**C= 21**✓
0	**D = 4**✓	**E = 340 kpa**✓	21

(5)

4.4
- To protect hands from arc rays that can cause injury to the skin
- To prevent the welding sparks from burning the skin
- To prevent the hot spatter from burning the skin (Any 2 × 1) (2)

4.5
- Failure to purge gas system
- Incorrect pressure on gas regulators
- Kinked hoses
- Overheated torch or nozzle
- Loose nozzles on torch (Any 3 × 1) (3)

[25]

QUESTION 5

5.1
- Welding is possible in all positions.
- Less operator skills are required.
- Long welds can be made without starting and stopping.
- Very little cleaning is required.
- Low cost. (5)

5.2
- Ensure that the correct PPE is worn.
- Check the safety signs in the area.
- Check for any hazards where the welding is to be done.
- Perform tests for any explosive gases.
- Ensure that the fire-fighting department or the fire extinguisher is close by.
- Ensure that the fire extinguisher is in good condition. (Any 5 × 1) (5)

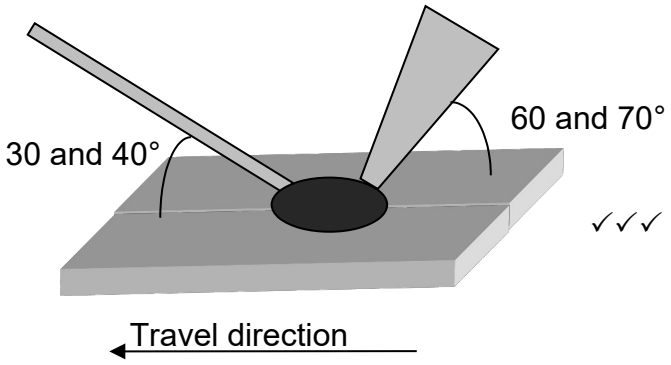
A – Regulator
B – Wire spool
C – Wire feeder
D – GMAW machine
E – Feeder gun/Electrode gun
F – Gas shield
G – Workpiece
H – Feeder gun cable
I – Earth cable
J – Shielding gas (10)
[20]

QUESTION 6

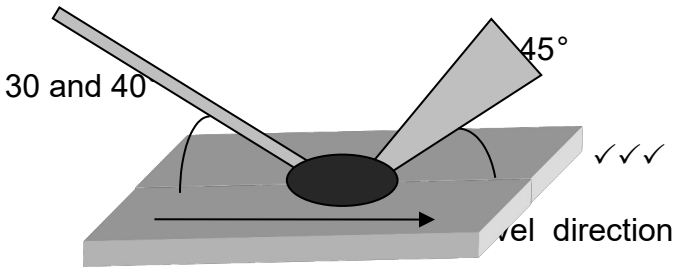
6.1.
- Ensure that the acetylene cylinder is securely chained.
- Acetylene cylinders must be stored separately from oxygen cylinders.
- Acetylene cylinders must not be stored near any heating areas. Example: blast furnace
- Ensure that the acetylene cylinders are stored in a properly ventilated area.
- Ensure that acetylene cylinders are stored away from sparks and open flames.
- Ensure that the acetylene cylinders are stored in an upright position.

(Any 4 × 1) (4)

6.2 6.2.1



6.2.2



(3 × 2) (6)
[10]

TOTAL: 100

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